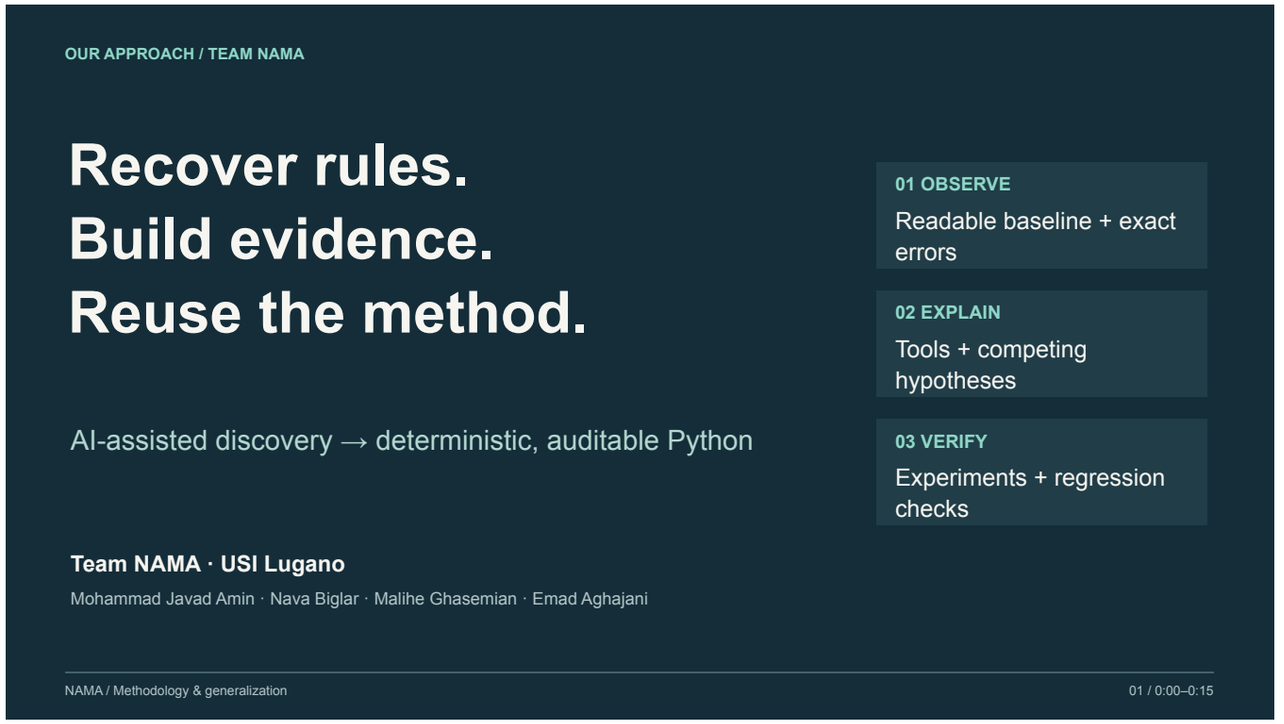
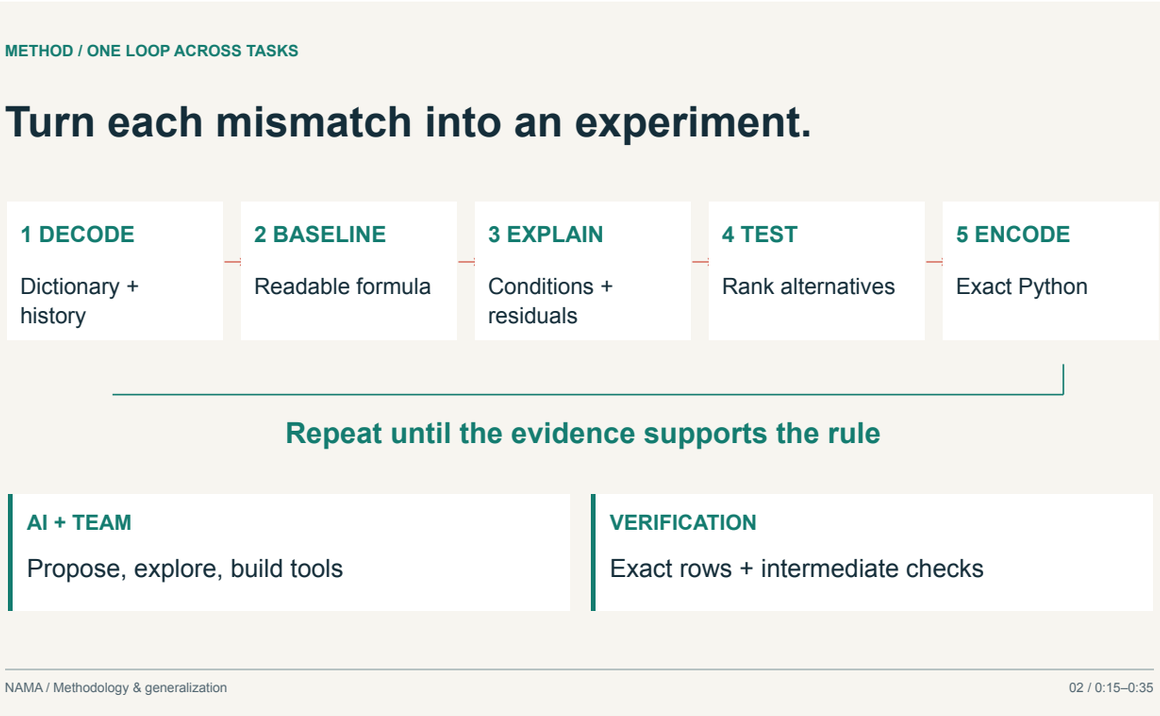


Recover rules. Build evidence. Reuse the method.
TIME: 0:00–0:15

SAY:
Our approach is a reusable discovery loop. We start with a readable rule, use the mismatches to guide experiments, and turn supported findings into deterministic Python. AI helps investigate; exact comparisons decide what we keep.

CUE:
Face camera; then bring up the three-step stack.

SUPPORT / EVIDENCE (not spoken):
Opening goes directly to our approach: assume the jury knows the challenge. Assistants used in development include Claude Code and Codex; the submitted calculations are deterministic. This is a human-guided toolkit, not autonomous recovery of arbitrary formulas.



Turn each mismatch into an experiment.
TIME: 0:15–0:35

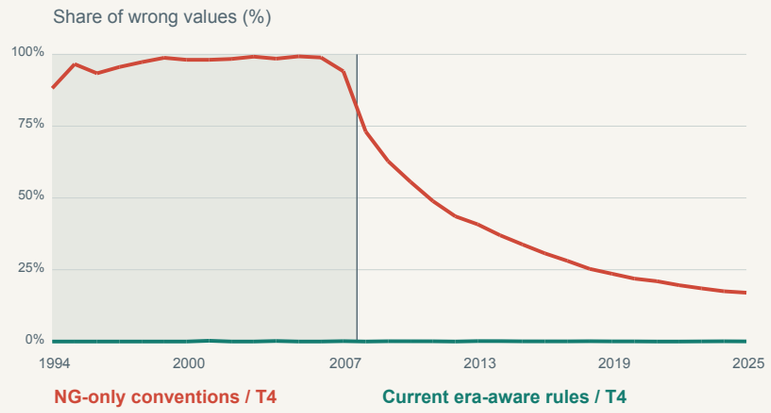
SAY:
A dictionary-driven loader gives us consistent inputs. We write a baseline, then ask where, when and how it fails. Trees identify conditions; change-point and residual tools suggest conventions. We test alternatives and rerun dependent tasks before integrating a rule.

CUE:
Trace the loop left to right; emphasize testing rather than hypothesis generation alone.

SUPPORT / EVIDENCE (not spoken):
Actual tools: legacy_loader.py, tools/rule_finder.py, tools/change_points.py, tools/residuals.py, tools/convention_search.py. The tree classifies mismatches; it does not establish the true formula. Convention search enumerates supplied candidate sets. Humans select features, propose candidates and assess evidence.

EXAMPLE 1 / ACTUAL REPOSITORY DIAGNOSTIC

Find where the system changed.



Local T4 training rows · NG-only: recorded era experiment · current: verified final snapshot

CLUE

Errors cluster by historical era.

EXPERIMENT

Test dated rounding, age and charge conventions.

Find where the system changed.
TIME: 0:35–0:58

SAY:
Applying newer conventions to older policies creates a strong historical pattern. Change-point analysis directs us to the conversion; we then test rounding, age and day-count candidates. The result is a dated rule with evidence. Both curves here show T4 training statements.

CUE:
Point to the older era, then to persistent errors after conversion. Explain carry-forward effects briefly.

SUPPORT / EVIDENCE (not spoken):
Source: methodology_sample/era_evidence.csv. Red is the recorded T4 Neuron NG-only experiment from Plots/errors_by_era.csv. Teal is recalculated from the verified latest T4 snapshot and all 34,145 local training rows, with21 mismatches. This is local training evidence, not private accuracy or a fresh NG-only ablation of the final candidate.
The annual plot motivates inspecting an era boundary; it does not independently identify the exact day. Current rules use conversion2007-08-11. Residual rates are small but not universally zero.

EXAMPLE 2 / CLEARLY LABELLED ILLUSTRATIVE ARITHMETIC

Use the error shape to propose a formula.

ILLUSTRATION · value 10,000 · annual rate 1% · period 31 days

CANDIDATE A · FIXED MONTH

10,000 × 1% ÷ 12
8.33

CANDIDATE B · ACTUAL/360

10,000 × 1% × 31 ÷ 360
8.61

Unrounded ratio: 31 / 30 = 1.0333…

Residual clue → candidate convention → exact comparison

The search reuses the same engine; each task supplies its candidate choices and scoring function.

Use the error shape to propose a formula.
TIME: 0:58–1:18

SAY:
Residuals also suggest formulas. This illustrative thirty-one-day charge is about three percent above a fixed monthly charge, suggesting actual-days over three-sixty. Our search tests competing day counts and rounding modes against real diagnostics. A plausible signature still needs exact validation.

CUE:
Point from the 1.033 ratio to the proposed day count; do not describe the toy amounts as measured outcomes.

SUPPORT / EVIDENCE (not spoken):
Illustration: value 10,000, annual rate 1%, elapsed period 31 days. Fixed month: 10,000×0.01/12=8.3333…; Actual/360: 10,000×0.01×31/360=8.6111…. Ratio of unrounded values=31/30=1.03333…. Displayed 8.33 and 8.61 use half-up cents for illustration; the historical era can use truncation. These are not observed policy outputs. Source tools: residuals.implied_ratios, convention_search.search, date_conventions.py.

EXAMPLE 3 / T7 TRANSFERS THE METHOD

Keep investigating when amounts are missing.

Observed witness · BL10002408 / event 8 · Swiss tax residence

GROSS / CHF	NET / CHF	LEGACY EVIDENCE
5,042.28 Above 5,000	4,941.43 Below 5,000	No VST booking

Supports a penalty/threshold hypothesis.

Booking presence constrains applicability. It cannot validate the amount or remittance date.

Sources: Version-2 investigation + current actual-data trace. T7 target amounts are unavailable locally.

Keep investigating when amounts are missing.
TIME: 1:18–1:38

SAY:
For T7, we reused the workflow with weaker evidence. This actual Swiss example crosses the threshold between gross and net proceeds, supporting investigation of penalties. Booking presence constrains applicability; it cannot establish the amount. Public examples add checks, while hidden accuracy remains unresolved.

CUE:
Use this as a generalization example; state the limitation in the same breath.

SUPPORT / EVIDENCE (not spoken):
Actual witness BL10002408/event 8, country CH, status E. research/t7_version2_experiments.md records gross tax-currency proceeds 5,042.28 and net 4,941.43; no VST booking. Latest cases.json confirms net CHF base 4,941.43 and no booked tax types. Policy currency is EUR, so label the displayed amounts CHF after conversion, not raw policy amounts. Booking absence supports applicability, not cent-exact tax amounts or remittance. Latest reported evaluated baseline: payout 1/20 and annual 15/20; current TRACE candidate has not been scored in this conversation.

GENERALIZATION / WHAT TRANSFERS

New target. Same investigative tools.

REUSE

Residual analysis
Trees + change points
Candidate search + exact checks

ADAPT

Schema + relationships
Candidate features and formulas
Reference evidence

99.735–100%

Exact rows · T0–T4 · local training targets

Human-guided discovery. Evidence-backed implementation.

T7 evaluated baseline: payout 1/20; annual 15/20. Current TRACE candidate: organizer result pending.

New target. Same investigative tools.
TIME: 1:38–1:55

SAY:
For another target, we supply new features, candidate formulas and evidence. We reuse the analysis, searches and validation process. Local T0-to-T4 scores exceed ninety-nine point seven percent. Our contribution is a reusable, human-guided method, with clear boundaries around verified results.

CUE:
Close on the transfer argument. Supporting slides are skipped during the two-minute talk.

SUPPORT / EVIDENCE (not spoken):
Local train scores from the full regression: T0 225848/225858, T1 15028/15028, T2 3011/3019, T3 6137/6153, T4 34124/34145. These are not private results. T7 last reported evaluated versions 1/20 payout and 15/20 annual; latest TRACE overlay is unscored. Generality is demonstrated across targets within this challenge, not across arbitrary unseen industries. Decoder formats and relationship browsers require adapters. Maintainability latest local82.5, wall/RSS overhead documented in t7_trace.md.

SUPPORTING SLIDE / APIS AND RESPONSIBILITIES

The reusable toolkit has explicit inputs.

Rule finder	features + mismatch flags	Readable conditions to investigate
Change points	dates + mismatch flags	Ranked historical boundaries
Residuals	calculated + reference values	Rounding and ratio signatures
Convention search	candidate options + score(choice)	Ranked candidate combinations
Weak labels	predicted applicability + bookings	Agreement / disagreement cases

Stack: Python · pandas · scikit-learn · matplotlib · Decimal

The reusable toolkit has explicit inputs.
TIME: APPENDIX
SUPPORTING SLIDE — skipped in the two-minute presentation.

SAY:
Use only for follow-up questions.

CUE:

SUPPORT / EVIDENCE (not spoken):
These are the current source APIs, not an invented autonomous agent architecture. Convention candidates and residual signatures include supplied insurance/date conventions, even though the search/ranking engine is reusable. LLMs accelerate development and hypothesis generation; exact comparison does not by itself prove unseen-data correctness.

SUPPORTING SLIDE / MEASURED RESULTS AND LIMITS

Keep local evidence separate from hidden scores.

LOCAL TRAINING TARGETS

T0 Commission	225,848 / 225,858	**99.9956%**
T1 Due dates	15,028 / 15,028	**100%**
T2 Maturity	3,011 / 3,019	**99.7350%**
T3 Surrender	6,137 / 6,153	**99.7400%**
T4 Statement	34,124 / 34,145	**99.9385%**

MAINTAINABILITY

82.5
Local scorer

REGRESSION

27 T7 tests
29 T6 checks

T5: 55 flags; hidden F1 unavailable. T7 evaluated baseline: 1/20 payout, 15/20 annual.

Latest TRACE overlay: seven stamp-only payout changes; all 6,381 annual rows unchanged vs updated main.

Keep local evidence separate from hidden scores.
TIME: APPENDIX
SUPPORTING SLIDE — skipped in the two-minute presentation.

SAY:
Use only for follow-up questions.

CUE:

SUPPORT / EVIDENCE (not spoken):
Scores copied from .cache/t7-trace-rebase/after.log. Denominators include the observed local mismatches; do not assert all unlisted misses are confirmed manual corrections. T5 is a set of55 flags, not a measured hidden F1. T6 has29 regression checks, which do not constitute a jury score. Snapshot comparison covers 6,782,289 booking digests and22,393 unit histories. T7 current overlay changes7 stamp values, preserves6381 annual rows, and passes27 synthetic tests. MI82.5 versus updated-main82.9. Three-run median T7 wall23.720 vs23.633s; peak RSS1779.55 vs1779.02MiB.

SUPPORTING SLIDE / DISCLOSURE AND EVIDENCE

Show what supported each decision.

OBSERVED

Training diagnostics, bookings, version history and organizer counts.

ILLUSTRATIVE

The charge arithmetic example explains a signature; it is not a policy trace.

AI DISCLOSURE

Claude Code + Codex assisted investigation, coding and review. The team directs and evaluates.

BOUNDARY

Reusable tools need new adapters and hypotheses. Hidden T7 accuracy remains unresolved.

Show what supported each decision.
TIME: APPENDIX
SUPPORTING SLIDE — skipped in the two-minute presentation.

SAY:
Use only for follow-up questions.

CUE:

SUPPORT / EVIDENCE (not spoken):
AI usage: Claude Code and Codex assisted coding, experiments, review, documentation and presentation preparation. Team members set direction, selected candidate hypotheses and assessed results. Record actual usage accurately; the existing report records organizer clarification on coding-assistant use for T7, which should be retained only if the team can substantiate it. No language model is invoked by the submitted task entry points. Evidence sources: challenge PDFs; tools/*.py; Plots/errors_by_era.csv; research/t7_version2_experiments.md; research/t7_trace.md; local diagnostic JSON and verification logs. The slide4 calculation is explicitly illustrative. Freeze final metrics against the actual submitted code before recording.